第三章　　罗弘纲　2021111778　车辆4班王.

3-1

3-2　$x_o(t)=t-0.1[1+e^{-0.5t}(3\sin 3.11t-\cos 3.11t)]$

单位阶跃响应:

$$x_o(t)=\left[t-0.1-0.1e^{-0.5t}(3\sin 3.11t-\cos 3.11t)\right]'$$

$$=1+0.1\times 0.5e^{-0.5t}(3\sin 3.11t-\cos 3.11t)-0.1e^{-0.5t}(9.33\cos 3.11t+3.11\sin 3.11t)$$

$$=1-e^{-0.5t}(0.1615\sin 3.11t+0.983\cos 3.11t).$$

单位脉冲响应

$$w(t)=\left[t-0.1-0.1e^{-0.5t}(3\sin 3.11t-\cos 3.11t)\right]''$$

$$=(1-e^{-0.5t}(0.1615\sin 3.11t+0.983\cos 3.11t))'$$

$$=0.5e^{-0.5t}(0.1615\sin 3.11t+0.983\cos 3.11t)-e^{-0.5t}(0.50071\cos 3.11t-3.05713\sin 3.11t).$$

$$=e^{-0.5t}(3.13763\sin 3.11t-0.00921\cos 3.11t).$$

3-3　$G_K(s)=\frac{4}{s(s+1)}=\frac{4}{s^2+s}$　$X_o(s)=G(s)X_i(s)=\frac{4}{s^2+s+4}\cdot\frac{1}{s}$　$\therefore \omega_n=2$　$\xi=\frac{1}{4}$

单位阶跃响应:　$\omega_d=\omega_n\sqrt{1-\xi^2}=\frac{\sqrt{15}}{2}$　欠阻尼系统

$$x_o(t)=L^{-1}[G(s)\cdot X_i(s)]=L^{-1}\left(\frac{4}{s^2+s+4}\cdot\frac{1}{s}\right)=1-\frac{4\sqrt{15}}{15}e^{-\frac{t}{2}}\sin\left(\frac{\sqrt{15}}{2}t+\arctan\sqrt{15}\right)$$

单位脉冲响应:

$$w(t)=L^{-1}[G(s)X_i(s)]=L^{-1}\left(\frac{4}{s^2+s+4}\right)=1-\frac{e^{-\frac{1}{2}t}}{\sqrt{1-\frac{1}{16}}}\sin\left(2\times\frac{\sqrt{15}}{4}t\right)=1-\frac{4\sqrt{15}}{15}e^{-\frac{t}{2}}\sin\frac{\sqrt{15}}{2}t.$$

3.4. (1) $g(t)=0.2e^{-0.4t}-e^{-0.1t}$

$$G(s)=0.2\left(\frac{1}{s+0.4}-\frac{1}{s+0.1}\right)$$

$$=\frac{-0.06}{(s+0.4)(s+0.1)}$$

$$=\frac{-0.06}{s^2+0.5s+0.04}$$

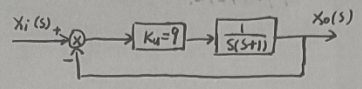
(2). $g(t)=0.5t+5\sin\left(3t+\frac{\pi}{3}\right)$

$$=0.5t+\frac{5}{2}\sin 3t+\frac{5\sqrt{3}}{2}\cos 3t$$

$$G(s)=\frac{1}{2s^2}+\frac{\frac{15}{2}}{s^2+9}+\frac{\frac{5\sqrt{3}}{2}s}{s^2+9}$$

$$=\frac{1}{2s^2}+\frac{5\sqrt{3}(\sqrt{3}+s)}{2s^2+18}$$

3-5. (1).



由图得 $G_K(s)=\frac{9}{s(s+1)}$

闭环传递函数 $G(s)=\frac{G_K}{1+G_K}=\frac{9}{s^2+s+9}$

$\therefore\ \omega_n=3 \quad \xi=\frac{1}{6}$ 二阶欠阻尼系统.

(2). $\omega_d=\omega_n\sqrt{1-\xi^2}=\frac{\sqrt{35}}{2}$

最大超调量 $M_p=e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}=e^{-\frac{\pi}{\sqrt{35}}}=0.588=58.8\%$

峰值时间 $t_p=\frac{\pi}{\omega_d}=\frac{2\pi}{\sqrt{35}}=1.062\,s$

调整时间 t_s 若 $\Delta=2\%$ $t_s=\frac{4}{\xi\omega_n}=8\,s$

若 $\Delta=5\%$ $t_s=\frac{3}{\xi\omega_n}=6\,s$

振荡次数 N 若 $\Delta=2\%$ $N=\frac{2\sqrt{1-\xi^2}}{\pi\xi}=\frac{2\sqrt{35}}{\pi}=3.77\approx4$ 次

若 $\Delta=5\%$ $N=\frac{1.5\sqrt{1-\xi^2}}{\pi\xi}=\frac{1.5\sqrt{35}}{\pi}=2.82\approx3$ 次

3-6 $\because$ 开环传递函数为

$$G_K(s)=\frac{1}{s(s+5)}$$

$\therefore$ 闭环传递函数为 $G(s)=\frac{G_K(s)}{1+G_K(s)}=\frac{1}{s^2+5s+1}$

$\therefore$ 该系统为二阶过阻尼系统

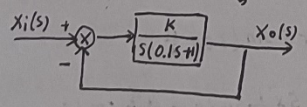
$\omega_n=1 \quad \xi=\frac{5}{2}$ 过阻尼系统无峰值时间和最大超调量.

$\therefore$ 上升时间 $t_r=\frac{1+1.5\xi+\xi^2}{\omega_n}=\frac{1+1.5\times\frac{5}{2}+(\frac{5}{2})^2}{1}=11\,s$

调整时间 t_s: 若 $\Delta=2\%$ $t_s=\frac{4}{\xi\omega_n}=1.6\,s$

若 $\Delta=5\%$ $t_s=\frac{3}{\xi\omega_n}=1.2\,s$

3-7.



$\therefore$ 开环传递函数 $G_K(s)=\frac{K}{s(0.1s+1)}$

闭环传递函数 $G(s)=\frac{G_K(s)}{1+G_K(s)}=\frac{\frac{K}{0.1s^2+s}}{1+\frac{K}{0.1s^2+s}}=\frac{10K}{s^2+10s+10K}$

$\omega_n^2=10K$ 且 $2\xi\omega_n=10$

$\because \xi=0.5 \quad \therefore K=10$

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3-8. 闭环传递函数 $G(s)=\dfrac{\frac{K}{s^2}}{1+\frac{K}{s^2}(1+K_f s)}=\dfrac{K}{s^2+K\cdot K_f s+K}$

∵ 峰值时间 $t_p=\dfrac{\pi}{\omega_d}=\dfrac{\pi}{\omega_n\sqrt{1-\xi^2}}=2s$

最大超调量 $M_p=e^{\frac{-\xi\pi}{\sqrt{1-\xi^2}}}\times 100\%=25\%$

联立方程 $\begin{cases}\omega_n=\dfrac{\pi}{2\sqrt{1-\xi^2}}\\ \dfrac{\xi}{\sqrt{1-\xi}}=\dfrac{2\ln 2}{\pi}\end{cases}$ 解方程得 $\begin{cases}\xi=0.4037\\ \omega_n=1.716\end{cases}$

∴ $K=\omega_n^2=2.945$

$K\cdot K_f=2\omega_n\xi=1.385$ $K_f=0.470$

3-9. $G_K(s)=\dfrac{\frac{K}{5}}{s(s+1)(\frac{s}{5}+1)}$ ∴该系统为1型系统 开环增益为 $K^*=\frac{K}{5}$

$e_{ss}=e_{ss}=\dfrac{1}{K^*}=\dfrac{5}{K}=0.01$ ∴ $K=500$.

3-10. (1) $G_K(s)=\dfrac{50}{(0.1s+1)(2s+1)}$

此系统为0型系统.

$K_p=\lim\limits_{s\to0}G_K(s)H(s)=\lim\limits_{s\to0}\dfrac{50}{(0.1s+1)(2s+1)}=50$

$K_v=\lim\limits_{s\to0}G_K(s)H(s)=\lim\limits_{s\to0}s\dfrac{50}{(0.1s+1)(2s+1)}=0$

$K_a=\lim\limits_{s\to0}G_K(s)H(s)=\lim\limits_{s\to0}s^2\dfrac{50}{(0.1s+1)(2s+1)}=0$

单位阶跃信号: $e_{ss}=\dfrac{1}{1+K_p}=\dfrac{1}{51}$

单位斜坡信号 $e_{ss}=\dfrac{1}{K_v}=\infty$

单位加速度信号. $e_{ss}=\dfrac{1}{K_a}=\infty$

(2). $G_K(s)=\dfrac{50}{s(0.1s+1)(2s+1)}$

此系统为1型系统

$K_p=\lim\limits_{s\to0}\dfrac{50}{s(0.1s+1)(2s+1)}=\infty$

$K_v=\lim\limits_{s\to0}\dfrac{50}{(0.1s+1)(2s+1)}=50$

$K_a=\lim\limits_{s\to0}\dfrac{s^2\ 50}{s(0.1s+1)(2s+1)}=0$

单位阶跃信号 $e_{ss}=\dfrac{1}{1+K_p}=0$

单位阶跃信号 $e_{ss}=\dfrac{1}{K_v}=\dfrac{1}{50}$

单位加速度信号 $e_{ss}=\dfrac{1}{K_a}=\infty$

(3) $G_K(s)=\dfrac{50}{s^2(0.1s+1)(2s+1)}$

此系统为2型系统

$K_p=\infty$ $K_v=\infty$ $K_a=50$

单位阶跃信号 $e_{ss}=0$

单位斜坡信号 $e_{ss}=0$

单位加速度信号 $e_{ss}=\dfrac{1}{50}$

(4) $G_K=\dfrac{50}{s^3(0.1s+1)(2s+1)}$

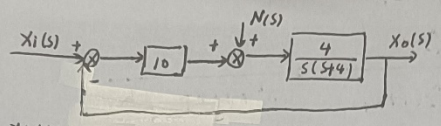
$K_p=\infty$ $K_v=\infty$ $K_a=\infty$

单位阶跃信号: $e_{ss}=0$

单位斜坡信号: $e_{ss}=0$

单位加速度信号: $e_{ss}=0$

3-11



$x_i(t) = t\,1(t) \qquad n(t) = 2\cdot n(t)$

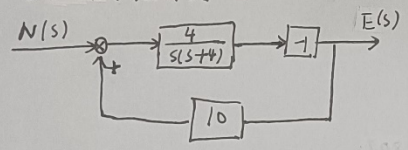
令 $n(t)$ 扰动为0时,

$$G_k(s) = \frac{40}{s(s+4)} = \frac{10}{s(0.25s+1)}$$

该系统为1型系统

$$e_{ssi} = \frac{1}{K} = 0.1$$

当 $N(s)$ 单独作用时

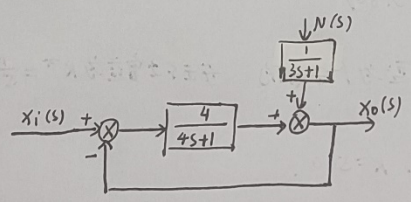


$$e_{ssd} = \lim_{s\to 0} s\,\frac{\frac{-4}{s(s+4)}\cdot 1\times\frac{2}{s}}{1+\frac{40}{s(s+4)}} = \frac{\frac{-8}{s(s+4)}}{\frac{s^2+4s+40}{s(s+4)}}$$

$$= \lim_{s\to 0}\frac{-8}{s^2+4s+40} = -0.2$$

$$\therefore e_{ss} = e_{ssi} + e_{ssd} = 0.1 - 0.2 = -0.1$$

3-12.



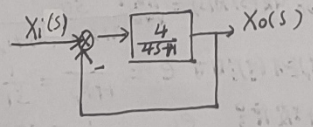
① 当 $X_i(s)=0 \quad N(s)\neq 0$ 时

$$X_{oN}(s) = \frac{1}{3s+1}\cdot\frac{1}{1-\frac{-4}{4s+1}}N(s) = \frac{4s+1}{(3s+1)(4s+5)}N(s)$$

$$E_N(s) = -X_{oN}(s) = -\frac{4s+1}{(3s+1)(4s+5)}N(s)$$

$$e_{ssN}(s) = \lim_{s\to 0} s\,E_N(s) = -\lim_{s\to 0} s\cdot\frac{4s+1}{(3s+1)(4s+5)}\cdot\frac{1}{s} = -\frac{1}{5}$$

② 再求当 $X_i(s)\neq 0$, $N(s)=0$ 时的稳态误差 e_{ssx}



$$\frac{X_o(s)}{X_i(s)} = \frac{4}{4s+5}$$

$$E_x(s) = X_i(s) - X_o(s) = \frac{4s+1}{4s+5}X_i(s)$$

$$e_{ssx}(s) = \lim_{s\to 0} s\,E_x(s) = \lim_{s\to 0} s\cdot\frac{4s+1}{4s+5}\cdot\frac{1}{s} = \frac{1}{5}$$

③ $e_{ss} = e_{ssx} + e_{ssN} = \frac{1}{5} - \frac{1}{5} = 0$